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Christopher Boerner

Electrical Engineer

Active Top Secret Clearance
Eagle Scout
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Electrical Engineer at Northrop Grumman supporting the Command & Data Handling (C&DH) subsystem on the ESPAStar satellite product line. Computer Engineering Master's graduate with hands-on experience spanning embedded C firmware, custom PCB design, and FPGA/SoC integration. Background includes hardware bring-up and validation, sensor integration, and digital logic implementation.

Skills

Languages & Software: C/C++, Python, Verilog, STOL, Bash, MATLAB, Xilinx Vivado/Vitis, STM32CubeIDE, KiCad, LTSpice, Cadence Virtuoso, Altium, Git, Linux

Circuit Design & EDA: PCB Layout, Schematic Capture, Signal Conditioning, Power Electronics, FPGA/SoC Integration, UART/SPI/I2C Interfacing

Instrumentation & Test: Oscilloscope, DMM, Logic Analyzer, Power Supplies, Scopy, PIL/HIL Validation

Education

Master of Engineering in Computer Engineering — May 2025
Virginia Tech – Focused on Computer Systems – GPA: 3.8 — *Alexandria, Virginia*
Advisor: Dr. Cindy Yi (Virginia Tech)

Bachelor of Science in Computer Engineering — May 2024
Virginia Tech – General Computer Engineering – GPA: 3.6 — *Blacksburg, Virginia*

Experience

Northrop Grumman | ESPAStar Product Line — Nov 2025 – Present
Electrical Engineer — *Dulles, Virginia*

- Support the Command & Data Handling (C&DH) subsystem on the ESPAStar product line – Northrop Grumman's ESPA-class satellite bus – and its hosted payloads.
- As a Systems Engineer on the NASA Gateway program, analyzed electrical schematics, pinouts, and block diagrams for the Electrical Power System to develop hardware verification test procedures.
- Wrote and executed test scripts (Python, NASA STOL) against physical flight computers and flatsat hardware; traced command/telemetry flows through flight software to verify subsystem behavior.

Virginia Tech | FPGA-Accelerated Echo-State Network (Master's Project) — Nov 2024 – May 2025
Graduate Researcher — *Alexandria, Virginia*

- Implemented a floating-point Echo-State Network and online RLS trainer in bare-metal C on a Xilinx Zynq SoC for 2x2 MIMO-OFDM wireless channel prediction.
- Integrated FPGA logic with the ARM processing system using Vivado/Vitis; achieved sub-1 ms inference latency with live Ethernet dataset streaming and UART monitoring.

Grenoble Electrical Engineering Laboratory | Expe-SmartHouse — Jun 2024 – Aug 2024
Research Intern — *Grenoble, France*

- Designed photovoltaic-powered smart home mockups with load management and real-time power monitoring.
- Programmed Arduino and ESP8266 microcontrollers for MQTT telemetry to a centralized Raspberry Pi energy manager.

Virginia Tech/NAVAIR | Aircraft Data Acquisition System (Senior Design) — Aug 2023 – May 2024
Team Member — *Blacksburg, Virginia*

- Designed custom PCBs with regulated power supplies, Wheatstone bridge signal conditioning, and optocoupler isolation for vibration, temperature, humidity, and acoustic sensor channels.
- Programmed STM32 and Artemis MCUs in C to acquire and packetize sensor data; transmitted over RF transceivers to a handheld unit for live telemetry display; validated under MIL-STD-810G.

Projects

Academic Projects

VLSI Design Project | 12-bit Multiplier in Cadence Virtuoso
Nov – Dec 2023

- Designed a 12-bit Braun multiplier (schematic/layout) with carry-select adders in Cadence Virtuoso.
- Verified via DRC/LVS/PEX; characterized propagation delay, power, and area for ADP optimization.

Integrated Design Project | Blood Oxygen Sensor
Jan – May 2022

- Created a multi-stage amplification and filtration circuit for photodiode sensor signals.
- Multiplexed red and infrared channels to calculate blood oxygen saturation.

Personal Projects

Smart Home Room Node | ESP32-S3, KiCad, Home Assistant
Mar – Present

- Designing a custom PCB room node integrating voice I/O, temperature/humidity sensing, and an e-ink display with LED ring.
- Full hardware cycle from breadboard prototype through KiCad PCB layout, SMD hot-air assembly, and 3D printed enclosure.

Distributed Homelab Infrastructure | TrueNAS, Docker, Linux
Jan – Present

- Self-hosted ZFS NAS, local AI stack (Whisper STT, Ollama LLM), and custom ESP32 room nodes.
- Deployed PoE-powered Raspberry Pi nodes for network-wide DNS filtering and Home Assistant smart home automation.